

UAVX PID Critique Report — Original Airframes

Date: 2026-08-08

Version: 4.0

Scope: Critique-only pass on `original/` airframes (legacy gains, not retuned) + v3.1 regeneration/validation of `user/_Tuned` + **v4 retirement & default-table sync** **Criteria:** FW_CRITERIA (attitude), AH_CRITERIA_FW (altitude hold), NAV_CRITERIA_FW (navigation)

v3.0 update: Re-run of the full critique. The simulation uses the **angle KP as regular proportional gains** ($\text{rate_desired} = \text{angle_error} \times \text{KP}$, exactly as in legacy `DoAngleControl`). This was already the case — the critique sim does **not** apply any quaternion $2 \cdot Q_a$ factor. (The quaternion cascade form only exists in the separate `test_quat_sim.py`.) Results below supersede v2.1.

v3.1 update (2026-08-06): The legacy `user/*_Tuned.af` fleet was deleted and **regenerated** as faithful copies of the `original/` airframes (correct AF_TYPE, no mislabeling). Fixed the simulator's `_Tuned`-suffix handling in `get_fw_descriptor` so a `user/X_Tuned.af` resolves its FW descriptor by stripping the suffix (previously `user/Phoenix_Tuned.af` etc. lost the aileron-less / rudderless structural gate and were wrongly scored — 17-20 FAILs). The MR frames carry their inherited legacy gains with a single defensible change (yaw-rate KP normalized toward default: `YAW_RATE_KP` = 0.25). Results below reconcile the regenerated fleet.

v4.0 update (2026-08-08): The `user/*_Tuned.af` fleet is **retired to** `airframes/backup/_retired_tuned/` — every tuned frame shipped the legacy raw `ROLL/PITCH/YAW_ANGLE_Q_INT_LIMIT` = 10 (10 rad/s) value in place of the intended $\approx 0.0026/0.0087$ rad/s conversion, so all of them fail `check_cascade_integrity` as a cascade-windup bomb (integrator above max commanded rate can never discharge). See `Cascade_Integrity_Checks.md`. Retune restarts from a clean baseline (`original/`). FC `params.c` ParamTable defaults for the Q angle loop are declared canonical and mirrored verbatim into GCS `PARAM_DEFAULTS` (tags 2/4/7/9/23/24/96/97/98): the GCS shipped 7.0/0.25 vs FC 1.75/0.0125 — divergence was 7-20 $\times$, now synced. Non-tune parameter limits (FC lo/hi ceilings + GCS `PARAM_LIMITS`) deliberately untouched (see `Airframe_Tuning_Normalization.md` §5).

v4.0 — Retire `user/_Tuned` fleet + default-table sync (2026-08-08)

The failure that stopped the retune

Running `check_cascade_integrity` (the cross-check from `Cascade_Integrity_Checks.md`) against `original/` (11 base frames incl. `user/Arado_555.af`, `user/Horten.af`) red-flags **every MR frame** with the same I-limit windup bomb:

```
original/Ecks_220mm.af : ! Yaw I-limit 10.00 > max commanded 3.67 - windup
original/Ken_450_1165.af: ! I-limit 28.00 > max commanded 7.33 - windup
...
generic/Dragon.af      : ! I-limit 10.00 > max commanded ... - windup
```

The value is not per-airframe-tuned; it is the legacy `uint8` 10 written straight into the unified-float field (3800 $\times$ over-budget). Every `_Tuned.af` inherited it, so no basis for per-frame retail was trustworthy. Hence the **decision to abandon the whole tuned fleet** and re-enter from the `original/` surface.

Decision and actions

1. **Retired** all 13 `user/*_Tuned.af` → `airframes/backup/_retired_tuned/` (the 2 unmatched `user/Arado_555.af` / `user/Horten.af` are kept in place as the return source).
2. **FC params.c defaults are the canonical Q-gain baseline** (AGENTS.md: FC is source of truth). The GCS `PARAM_DEFAULTS` for the Q angle loop now mirrors FC exactly:

tag	param	FC default	GCS default (after)	was (before)
2/7	Roll/PitchAngleQKp	1.75	1.75	7.0
4/9	Roll/PitchAngleQIntLimit	0.01	0.01	0.0026
23/24	Roll/PitchAngleQKi	0.0125	0.0125	0.25
96	YawAngleQKp	0.75	0.75	3.0
97	YawAngleQKi	0.0125	0.0125	1.0
98	YawAngleQIntLimit	0.03	0.03	0.0087

Note these are *schema* defaults only (UseDefaultParameters fresh-provisioning); they do not override the `.af` files' live flight values (unit-spinbox path). Non-tune limits unchanged (FC `PARAMLimits` / GCS `PARAM_LIMITS` stay where the check `test_param_limits.py` validates them).

3. **Next:** regenerate the tuned fleet from a cascade-corrected baseline and re-run the RIG until 0 hard fails.

Simulation convention (v3.0)

The attitude loop in `test_pid_sim.py` is the legacy angle-cascade:

```
run_angle_loop:  pi.Error = clamp(stick*Max + NavCorr) - angle
                  rate_desired = pi.Error * ANGLE_KP + ITerm          # regular KP, rad/s per rad
run_rate_pd:     out = clamp((rate_desired - rate) * RATE_KP + DTerm, -1, 1)
```

- `ANGLE_KP` is used **directly** (rate command per rad of angle error) — no $2 \cdot Q_a$ half-angle scaling, no quaternion-to-Euler conversion. A 10° error with `ANGLE_KP=7` commands ~ 1.22 rad/s.
- This matches legacy `UAVXArm32F4/src/control.c DoAngleControl: P->PTerm = P->Error * P->Kp → R.Desired = PTerm + ITerm → ControlRate (R.Kp in series)`.
- For small angles the quaternion controller's $2 \cdot Q_a \cdot QGain \approx \text{angle_error} \cdot QGain$, so treating angle KP as regular gains is the correct interpretation for tuning; it does **not** under-size authority.

Fixed Wing — Original (1 frame)

original/Phoenix.af

Result (v3.0): PASS

Test	Result	Details
Pitch @ 30°	PASS	—
Yaw @ 15°	PASS	—
Pitch disturbance rejection	PASS	—
Yaw disturbance rejection	PASS	—

Test	Result	Details
Alt Hold 5m	PASS	—
Nav 10m	PASS	—

Note: aileron-less RudderElevatorAF — roll step is a structural gate (non-scored).

Multicopter — Original (8 frames)

original/Ecks_220mm.af — FAIL

- Roll @ 15°: PASS · Pitch @ 10°: PASS · Yaw @ 45°: **ISSUES** (OS 17.5%, settle 8s)
- Roll disturbance: **ISSUES** (peak 17.8°/s) · Pitch disturbance: PASS · Yaw: **ISSUES** (peak 17.2°/s)
- AH 5m: **ISSUES** (rise 2.03s) · Nav 10m: PASS

original/Ken_450_1165.af — FAIL

- Roll @ 15°: PASS · Pitch @ 10°: PASS · Yaw @ 45°: **ISSUES** (settle 8s)
- Roll disturbance: **ISSUES** (peak 16.2°/s) · Pitch disturbance: PASS · Yaw: **ISSUES** (peak 13.4°/s)
- AH 5m: **ISSUES** (rise 2.33s) · Nav 10m: PASS

original/Ken_Alpha_Test.af — FAIL

- Roll @ 15°: PASS · Pitch @ 10°: PASS · Yaw @ 45°: **ISSUES** (settle 8s)
- Roll disturbance: **ISSUES** (peak 19.3°/s) · Pitch disturbance: PASS · Yaw: **ISSUES** (peak 16.0°/s)
- AH 5m: **ISSUES** (rise 3.34s) · Nav 10m: PASS

original/Ken_LadyBug.af — FAIL

- Roll/Pitch @ step: **ISSUES** (sim diverges — Roll final=NaN, peak=inf)
- Yaw @ 45°: **ISSUES**
- Roll/Yaw disturbance: **ISSUES** · Pitch disturbance: PASS
- AH 5m: **ISSUES** · Nav 10m: PASS
- **Note:** sim physics diverges on this frame (NaN/inf on roll step) — gains need review.

original/Ming_DevEBox.af — FAIL

- Roll @ 15°: PASS · Pitch @ 10°: PASS · Yaw @ 45°: **ISSUES** (settle 8s)
- Roll disturbance: **ISSUES** (peak 19.3°/s) · Pitch disturbance: PASS · Yaw: **ISSUES** (peak 16.0°/s)
- AH 5m: **ISSUES** (rise 2.09s) · Nav 10m: PASS

original/S500_1137.af — FAIL

- Roll @ 15°: PASS · Pitch @ 10°: PASS · Yaw @ 45°: **ISSUES** (settle 8s, OS 10.2%)
- Roll disturbance: **ISSUES** (peak 17.8°/s) · Pitch disturbance: PASS · Yaw: **ISSUES**
- AH 5m: **ISSUES** · Nav 10m: PASS

original/WLToys_LadyBug.af — FAIL

- Roll/Pitch @ step: **ISSUES** (same NaN/inf sim divergence as Ken_LadyBug — same gains/physicals family)
- Yaw @ 45°: **ISSUES**
- Roll/Yaw disturbance: **ISSUES** · Pitch disturbance: PASS
- AH 5m: **ISSUES** · Nav 10m: PASS

original/Rok_Quad.af — FAIL

- Roll @ 15°: PASS · Pitch @ 10°: PASS · Yaw @ 45°: **ISSUES** (settle 8s)
- Roll disturbance: **ISSUES** (peak 16.0°/s) · Pitch disturbance: PASS · Yaw: **ISSUES** (peak 16°/s)
- AH 5m: **ISSUES** (rise 3.79s) · Nav 10m: PASS
- **Note:** physicals updated (AUW=1344g, arm=260mm, prop=9x4, 5000mAh 3S, thrust=610g); gains not retuned.

Summary Table

Airframe	Category	Attitude	AH	Nav	Notes
Phoenix	FW	PASS	PASS	PASS	Roll structurally non-scored
Ecks_220mm	MR	Partial	ISSUES	PASS	Yaw settles 8s; roll/yaw gusts high
Ken_450_1165	MR	Partial	ISSUES	PASS	Yaw settles 8s
Ken_Alpha_Test	MR	Partial	ISSUES	PASS	Yaw settles 8s
Ken_LadyBug	MR	FAIL	ISSUES	PASS	Roll sim diverges (NaN)
Ming_DevEBox	MR	Partial	ISSUES	PASS	Yaw settles 8s
S500_1137	MR	Partial	ISSUES	PASS	Yaw settles 8s
WLToys_LadyBug	MR	FAIL	ISSUES	PASS	Roll sim diverges (NaN)
Rok_Quad	MR	Partial	ISSUES	PASS	Yaw settles 8s

Common failure pattern: all MR frames pass attitude roll/pitch steps and Nav, but fail on **yaw settle time** (8s vs 5s limit), **roll/yaw gust peak deviation** (16–19°/s vs 12–15°/s limits), and **AH rise time** (2.0–3.8s vs 2s limit). These point at under-gained yaw-rate and altitude-ROC loops in the legacy gains, not the angle KP scaling (which is regular and correct).

Recommendations

1. **Do not deploy original/ airframes with legacy gains** — they fail yaw/AH criteria.
 2. **If any original airframe is to be used**, run the full tuning pipeline (attitude + AH + Nav) as done for generic/ frames — yaw KP/KI and alt-ROC KP are the primary targets.
 3. **Ken_LadyBug / WLToys_LadyBug** — sim physics diverges on roll step (NaN/inf); gains and physicals need review before they can be tuned.
 4. **Rok_Quad** — physicals updated; needs full retune for the new mass/prop configuration.
 5. **Phoenix** — passes with the v2.1 retune; unchanged in v3.0.
-

Verification

Run any original airframe critique:

```
python3 -m test_pid_sim original/Rok_Quad.af
python3 -m test_pid_sim original/Phoenix.af
```

Expected: PASS for Phoenix; FAIL/ISSUES on yaw, roll/yaw gust, and AH rise for the MR frames (Nav now passes).

v3.1 — Regenerated user/*_Tuned.af fleet (2026-08-06)

The active tree previously carried mislabeled user/_Tuned frames (e.g. Radian_Tuned.af, Arado_555_Tuned.af were QUAD_X MR configs under FW names — these would select the wrong mixer/motor map and crash on spin-up). They were **deleted**, then **regenerated** as faithful copies of the original/ airframes with their true AF_TYPE, so each is a valid read-write tuning baseline (Radian_Tuned → RUDDER_ELEVATOR, Arado_555_Tuned → ELEVON).

Simulator fix

test_pid_sim.py::get_fw_descriptor and is_yaw_structurally_limited now strip a _Tuned.af suffix before the basename match. Without this, a regenerated FW tuned copy failed to resolve its aileron-less / rudderless descriptor, so the structural gates (is_roll_structurally_limited / is_yaw_structurally_limited) did not fire and the frame was scored with strict roll/overhead limits (max 10% overshoot) it could never meet, yielding 17-20 FALSE FAILs for Phoenix_Tuned, Radian_Tuned, Arado_555_Tuned.

user/Rok_Quad_Tuned.af was additionally mapped to QUAD_X (matching original/Rok_Quad.af); the earlier QUAD mapping triggered AF_CATEGORY KeyError → no results.

Validation (default gains, python3 tests/test_pid_sim.py <file>, no slider)

File	FAIL / PASS	Notes
user/Phoenix_Tuned.af	0 / 45	PASS — retuned baseline
user/Radian_Tuned.af	0 / 45	PASS — retuned baseline
user/Arado_555_Tuned.af	1 / 44	1 artifact: free-flight roll settle (see below)
user/Ecks_220mm_Tuned.af	3 / 42	MW yaw / roll-gust / AH (inherited legacy)

File	FAIL / PASS	Notes
user/Ken_450_1165_Tuned.41	FAIL	MW yaw / roll-gust / AH
user/Ken_Alpha_Test_Tuned.42	FAIL	MW yaw / roll-gust / AH
user/Ming_DevEBox_Tuned.42	FAIL	MW yaw / roll-gust / AH
user/S500_1137_Tuned.3f / 42	FAIL	MW yaw / roll-gust / AH
user/Rok_Quad_Tuned.a4 / 41	FAIL	MW yaw / roll-gust / AH
user/Ken_LadyBug_Tuned.7a36	FAIL	brushed-micro limits (see below)
user/WLToys_LadyBug_TB7e35af	FAIL	brushed-micro limits (see below)

Notes

1. **FW frames PASS after the descriptor fix** — the earlier 17–20 FAILs were classifier artefacts, not real gain problems. Their tuning carries over from the original baseline.
2. **Arado_555_Tuned single FAIL** is a free-flight (no PID) roll *settle* report: turbulence keeps the already-small peak ($\sim 2.8^\circ$) above 5% of its own peak within the 15s window, so the “-settle 5%” metric never triggers. It is a measurement/sampling artefact, not a gain deficiency — `PHYS_ROLL_DAMP` / `PHYS_DIHEDRAL` variants barely move it (7.8–7.1s across an $8\times$ damping change). Arado’s real geometry (45° sweep, inboard $+7^\circ$ dihedral, outboard -5° anhedral $\rightarrow$ effective dihedral small, sweep-induced adverse yaw large) is consistent with the stored `PHYS_DIHEDRAL=0.05` / `PHYS_ADVERSE_YAW=0.05`.
3. **MR micro frames (Ken_LadyBug / WLToys_LadyBug)** retain the brushed-motor divide: overshoot on roll/pitch step, yaw/roll-gust peak deviation, and slow AH. Raising rate/angle gains did not clear them (brushed-actuator + low inertia limit), so they are documented as physical — their original frames show the same pattern (see multicopter section above).
4. Yaw-rate for the `MR_Tuned` copies was normalized to `YAW_RATE_KP = 0.25` (default mid; was ~ 0.067), a single defensible change. `YAW_ANGLE_Q_KP` was left at the source value (2.25–3).